

THE COMPANION PROBLEM, FROM HARMONIC SUMS TO MODULAR ANCHORS: A PROGRAMME REPORT

ABSTRACT. This is the narrative report of a three-day campaign (2026-08-03–05) of the odd-zeta programme on the *companion problem*: closed formulas for the second solutions of Apéry-like recurrences — the fifteen sporadic pairs, and the rows of the Brown–Zudilin cellular family for $\zeta(5)$. The campaign began as a siege of one identity (the programme’s “bridge”), produced a wall of sharply structured negatives, extracted from those negatives first a proof-shaped obstruction (curve-blindness), then an instrument (the nome/anchor calculus), and ended with a proved theory: an abstract companion inversion theorem; a proof that all fifteen sporadic operators are exactly rectified by their nomes — for the six third-order families via explicit symmetric-square identities with a uniform closed form for the square root — and hence a uniform, all- n Eichler-integral companion formula for every family, including the seven for which the complete tame harmonic ansatz is (precisely, finitely) excluded. Along the way: proved modular identifications, a new uniform weight-one congruence law that survives sixteen held-out primes per family, a proved annihilation law for a “formula-less” family’s summand, a census finding no sixteenth sporadic pair in a defined search universe, mirror-type signatures of the $\zeta(5)$ operator, one refuted conjecture (reported with pride), and one small delightful object (the half-Apéry numbers) whose integrality is honestly trivial and whose Lucas law sees the class field of $\mathbb{Q}(\sqrt{-6})$. We narrate the campaign in the order it happened, because the logic of discovery — negatives with structure pointing at the missing instrument — is the transferable lesson. Every claim carries its evidence label and a pointer to its technical home; this survey asserts nothing above its source’s grade.

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1. THE PROBLEM, AND THE DISCIPLINE

1.1. **Companions.** An *Apéry-like pair* is a three-term recurrence with two distinguished solutions: an integral one $A(n)$, given by a binomial sum, and a *companion* $B(n)$ (normalised $B(0) = 0$, $B(1) = 1$), whose arithmetic — denominators, congruences, limits — carries the irrationality-theoretic content. Apéry’s $\zeta(3)$ proof is the archetype: $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, and the companion with $b_n/a_n \rightarrow \zeta(3)$, whose *closed formula* (a binomial sum against a harmonic weight) is what makes the denominator bound $d_n^3 b_n \in \mathbb{Z}$ provable. The fifteen sporadic pairs — Zagier’s six, the

six Almkvist–Zudilin cases, Cooper’s three — generalise the archetype; for eight of them harmonic companion formulas were known, and for seven (“the conjectural seven”: $\mathbf{B}, \mathbf{C}, \mathbf{F}, \delta, \zeta, \eta, s_{18}$) none was known, with mounting evidence of absence.

One floor up sits the Brown–Zudilin cellular family for $\zeta(5)$: a certified third-order recurrence L_{BZ} with rows $(Q_n, \hat{P}_n, P_n)$ and $I_n = Q_n(2\zeta(5) + 4\zeta(3)\zeta(2)) - 4\hat{P}_n\zeta(2) - 2P_n$. The middle row has a proved companion formula; the top row’s compact formula

$$P_n \stackrel{?}{=} \sum_{k,l=0}^n T(n, k, l) \omega_5(n, k, l), \quad T = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}, \quad (\text{BRIDGE})$$

is verified far beyond reasonable doubt and [OPEN]; it is the single identity on which the programme’s sharp low-prime denominator theorem for P_n (the factor $2^2 \cdot 3$) currently rests. Sharper proved denominators are the only known lever on the worthiness exponent of the $\zeta(5)$ approximations — which is why one identity gets a siege.

1.2. The discipline. The programme’s evidence classes — [THM]/[PROVED], [VERIFIED: range], [EXCLUDED: scope], [OPEN], and (used once below, with satisfaction) [REFUTED] — are load-bearing and never silently upgraded. Two standing rules were adopted mid-campaign at a referee’s insistence and are now programme law: finite coefficient agreement is never called an identity, and finite integrality is never called modularity. This survey cites four technical papers — [MA] (`modular_anchors`, the theory, post-referee), [VA] (`variational_anchor`, the jet calculus), [EA] (`expository_apery`), [HA] (`half_apery`) — and the memos and scripts catalogued in Appendix A.

2. ACT I: THE SIEGES, AND WHAT THEIR FAILURE TAUGHT

The campaign opened as an assault on (BRIDGE) through its ε -deformation reduction: the bridge follows (in part) from a single cellwise identity $\sum T\Delta_5 = 0$, whose symmetrised form had long been known to lie in the per-row kernel of the cell — and to sit *exactly one dimension* outside the span of every proved family of residue-calculus generators.

Six generator classes were built or completed, each calibrated (every generator individually certified as an identity) at two unrelated 22-bit primes:

- (1) fixed-pole jets, evaluation and derivative towers, and the anti-diagonal Laurent family (the programme’s prior alphabet, re-based);
- (2) elementary-symmetric pole cancellations on the simple-zero endpoint ranges and the full numerator multiset (rank $424 \rightarrow 426$);
- (3) z -weighted jets — valid identities living outside the pure harmonic ring, requiring an extended ring with polynomial letters (2030 generators; rank 1497);
- (4) z^2 -weighted jets and z -weighted towers (2429 generators; rank 2714);
- (5) nested harmonic-weighted jets, lattice and endpoint, with nested letters (258 generators);
- (6) and, on the telescoping side, capped order-zero certificates for the residual (excluded at depths 1–3, three denominator families, zero nullity at depth 3).

The target remained exactly one dimension outside every span [EXCLUDED: two primes, identical ranks; scripts `eps26–eps32`, `t_deep_r`]. Meanwhile the membership fact itself was stress-tested and held: the per-row kernel rank saturates at exactly 1673 out to $n \leq 75$ [VERIFIED: `eps33`] — a per-row decomposition exists; the constructive alphabet simply cannot reach it.

Two consolidations made the impasse precise. The missing dimension was extracted as one explicit 231-monomial identity with exact rational coefficients (two-prime CRT, matching pivot patterns; `eps31`) — the narrowest formulation of the gap to date. And capped order-zero telescoping certificates for that residual were excluded at inflation depths 1–3, three denominator families, with zero nullity at depth 3 (`t_deep_r`).

Lesson 2.1. Six independent local mechanisms missing by exactly one dimension is not bad luck; it is a signature. All the generators were *local* functionals of one kernel. The missing direction is a global invariant no local instrument detects. The productive move was not a seventh local family but the question: *what class of instruments are we actually using, and what is its reach?*

3. ACT II: CURVE-BLINDNESS, AND THE TOOL THAT FELL OUT

The reach question has a clean answer for the analytic instruments. A *curve atom* is a deformation of the cell along a polynomial curve $T(n + \alpha(\varepsilon), k + \beta(\varepsilon), l + \gamma(\varepsilon))$ — the meromorphically honest deformations, each a Brown–Zudilin parametric integrand at shifted parameters.

Lemma 3.1 (coupling; three lines, degree-independent). [PROVED] ([VA]) *A harmonic monomial of weight w at ε -order r needs letter-factors with $m_i \geq j_i$, $\sum m_i = r$, $\sum j_i = w$; at top grade $w = r$ this forces $m_i = j_i$ and $[\varepsilon^j]s_L^j = d_{1,L}^j$. So the top-graded content of $[\varepsilon^r]$ of any polynomial curve, of any degree, depends only on its first-order direction, through pure r -th powers. Higher curve coefficients feed only lower weights; and for mixtures, the fifth-symmetric moment slot cannot be steered independently of the lower moments (polarization is blocked — both move at the same order).*

Lemma 3.2 (Veronese saturation). [PROVED] (exact ranks 6, 10, 15, 21; [MA]) *The direction box $\{-2..2\}^3$ has full-rank Veronese images at degrees 2–5; the finite computations therefore certify the top grade for every integer direction, hence every polynomial curve.*

Theorem 3.3 (curve-blindness). ([MA], [VA]) *The top-graded ε^5 content of any pinned combination of polynomial-curve atoms has vanishing P_n -component — unconditionally, by Lemmas 3.1–3.2; and on the stated boxes the entire ε^5 row is confined to $\mathbb{Q}Q \oplus \mathbb{Q}\hat{P}$ [EXCLUDED: two primes; eps41/42]. The ε^3 row meanwhile realises $\hat{P}$ freely: analytic curve deformations see the $\zeta(3)$ level and are structurally blind to the $\zeta(5)$ level — matching the family’s weight spike $\{0, 3, 5\}$.*

The same fact appears on the formal side: the one known P -emitting deformation (“family 1”) has letter data that is not curve-consistent, and admits no Γ -product realisation at any reasonable shift scale (minimal: $\varepsilon/120$ with multiplicities $\sim 10^{11}$) [VERIFIED: eps40]. Blindness and non-realizability are one fact seen from two sides: *the top graded piece requires jet data decoupled from lower jets.*

3.1. The tool. Decoupled jets exist analytically: mixed parameter-derivatives of the cellular integral. What they need is an evaluation anchor, and its first rung was tested and passed:

$$L_{\text{BZ}}(\dot{Q})(n) = - \sum_{i=0}^3 c'_i(n) Q(n+i), \quad \dot{Q} = \sum_{k,l} T \cdot \Lambda_1^{(n)},$$

exactly [VERIFIED: exact $\mathbb{Q}$, $n \leq 18$; eps46], with a purely discrete proof route (differentiate the rational certificate; the k - and l -variations vanish identically — they are gauge). The tower has an honest rung-two obstruction at the sum level (continuation cells revive under second derivatives; seven of nine letter directions are safe, $n-k, n-l$ are not, and the B_5 weight as tabulated needs order-4 unsafe jets), with two identified escapes: a null representative of B_5 with safe orders, or the integral-level tower licensed by Brown–Zudilin’s own transfer principle, for which the explicit continuous-parameter contiguity matrices in the appendix of Brown’s Mellin-transform paper are the template ([VA]; memos VARIATIONAL_TOWER.md, BROWN26_MINING.md).

Lesson 3.4. The proof of an obstruction can name the missing instrument. Here it named two: polarized (decoupled-jet) objects for the $\zeta(5)$ row, and — via the question “what *does* the jet mechanism see?” — the discovery instrument of the next act.

4. ACT III: THE DISCOVERY INSTRUMENT, ITS NEGATIVES, AND A REFUTATION

Run in reverse (scan $\rightarrow$ rational reconstruction $\rightarrow$ exact held-out validation, always with a positive control), the jet calculus is a companion-discovery instrument.

Controls. On Franel and **D** — known companions — the pinned scan rediscovers reachability from linear atoms, minimal support, and the reconstructed weights verify exactly over $\mathbb{Q}$ on held-out ranges; they differ from the published formulas by proved-null weights, i.e. new equivalent representations [VERIFIED: exact $\mathbb{Q}$, $n \leq 25$; eps43]. (The control also caught a genuine int64 overflow — the standing argument for controls.)

Negatives with structure. The three families for which the sporadic status paper had proved no fit exists (**B**, δ , ζ) are jet-unreachable [EXCLUDED: two primes, linear+quadratic curves, ε^2 and ε^3 , every ζ -grade]. The paper’s three never-run full-degree fits were then run — families ζ , η at full degree, **F** at its seven tame arguments, pure and character-twisted — and all six are inconsistent, with excess 90–173 [EXCLUDED: eps44; precise ansatz definition and the modular-to-rational transfer lemma in [MA]]. The complete tame degree- ≤ 3 ansatz is thereby exhausted for all seven conjectural families. Two diagnostic corrections emerged: ζ *does* have summand-letter $\mathbb{Q}$ -relations at full degree (a 15-dimensional kernel; the old “rank = columns” claim fails for ζ), while η is maximally rigid even at 2530 columns.

The kernel pays. Extracting ζ ’s kernel produced two per-cell annihilation laws; the first is now a theorem:

$$\sum_l S_\zeta(n, k, l) (H_k^{(1)} - 2H_l^{(1)} + H_{n-l}^{(1)}) = 0 \quad (0 \leq k \leq n) \quad [\text{PROVED}] \quad (1)$$

by an order-2 Zeilberger certificate with symbolically verified certificate identities, endpoint vanishing, and unconditional induction (eps45; stated as a Proposition in [MA]) — the first proved structural fact about any conjectural-seven family’s letters. The second law remains [VERIFIED: $n \leq 25$] with a documented obstruction record [OPEN]. Moral: every fit for ζ had been conducted in a provably degenerate alphabet.

A refutation, reported with pride. The first form of the reachability dichotomy — jet-reachable $\iff$ principal character with real limit — made a prediction; its designed discriminating experiment (the Domb family α , and ε : principal character, real limits, *and* known harmonic companions) returned unreachable at both primes [REFUTED] (eps54). The surviving invariant is the *level of the uniformization*: squarefree (5, 6) reachable; non-squarefree (8, 9, 12, 12, 36) not — seven data points, no exceptions — and jet-reachability is strictly finer than formula existence: the jets detect the arithmetic of the uniformization, not the letters of the formula. The refuted conjecture is kept on the record in both papers.

Lesson 4.1. The campaign’s conjectures are required to make predictions, and its experiments are designed to kill them. One died. Its successor is better. This is the methodology working, not failing.

5. ACT IV: MODULAR ANCHORS — THE CENTERPIECE

The refuted conjecture’s successor pointed at uniformizations, so the campaign built the instrument that computes them: from a recurrence, the Frobenius basis y_0 , $y_1 = y_0 \log t + g$, the nome $q = t \exp(g/y_0)$, the mirror map $t(q)$, and $F(q) = y_0(t(q))$ — with the Apéry pair as validation (the instrument reproduces Beukers’ classical $\Gamma_0(6)$ eta parametrization exactly; [EA] walks a reader through this case with tables and code).

5.1. Identifications. All fifteen nome expansions are integral at scale one, and twelve of fifteen (t, F) pairs match explicit (generalized-)eta objects to q^{25} [VERIFIED: eps51; dictionary memo]. Statuses, per the standing rules:

- ζ : [PROVED]. $F_\zeta = \eta_3^{10}/(\eta_1\eta_9)^3 = \frac{1}{8}(9E_2(q^9) - E_2(q))$ is an identity in $M_2(\Gamma_0(9))$ (Ligozat conditions; Sturm bound 2; agreement to q^{40}) ([MA]).
- **B**: $F_B = b(-q)$, the Borwein cubic theta $\eta(q)^3/\eta(q^3)$ at $-q$ [VERIFIED: all 27 computed coefficients], with a level-36 half-integral shift anomaly and an equivalent untwisted level-9 representative found by the census.
- δ : level 12 eta quotients for both t and F — apparently the first modular data ever attached to δ — with the identification proved *modulo one hypothesis* (the degree-one property of t_δ on the correct Atkin–Lehner extension); the eta-side annihilator is exactly one-dimensional and equals δ 's operator coefficientwise (DELTA_MODULAR_THEOREM.md).
- η : $[\eta(q)^5/\eta(q^5)](-q)$, level 20 [VERIFIED: all 27 coefficients]. Cooper's three: integral, not eta-type in this coordinate [OPEN].

5.2. The theory: inversion, rectification, and the all- n companion formula. The centerpiece is now proved end to end ([MA]):

Theorem 5.1 (companion inversion). [THM] *If an order- r operator is exactly rectified by its nome, $L = C(t) F \theta_q^r F^{-1}$, then the normalized companion (the unique log-free solution of $L(y_B) = t$ vanishing at 0 — the boundary defect equation, itself a two-line lemma) is*

$$y_B = F \theta_q^{-r} \left(\frac{t}{C F} \right).$$

Theorem 5.2 (rectification for the fifteen; the six symmetric squares). [PROVED] *Order-2 operators are rectified by Frobenius theory (nine families, Cooper's $d \neq 0$ rows included). Each of the six third-order sporadic operators equals $t^3 P_3(t)$ times the symmetric square of*

$$\tilde{D} = \theta^2 - t \left(2a\theta^2 + a\theta + \frac{b}{2} \right) + c t^2 \left(\theta + \frac{1}{2} \right)^2,$$

uniformly in Zagier's parameters (a, b, c) — exact residual-zero identities in $\mathbb{Q}(t)$, family by family (eps57). (Rectification at $r \geq 3$ is strictly stronger than maximal unipotent monodromy; the referee's insistence on this distinction, and the resulting repair, are part of the record.)

Theorem 5.3 (uniform companion formula, all n). [THM] *For every one of the fifteen sporadic pairs,*

$$B(n) = [t^n] F \cdot \theta_q^{-r} \left(\frac{t \sigma^r}{P_r F} \right) \quad \text{for all } n \geq 0.$$

For the conjectural seven, whose tame harmonic ansatz is exhausted (§4), this is a companion formula in provably-necessary new letters — the first of any kind. Displayed for the family that started it:

$$B_\zeta(n) = [t^n] F_\zeta \cdot \theta_q^{-3} \left(\frac{t \sigma^3}{P_3 F_\zeta} \right), \quad F_\zeta = \frac{1}{8}(9E_2(q^9) - E_2(q)).$$

The companion problem was never about cleverer harmonic weights; it was about whether the family's uniformization can be rationally eliminated. Where it can, the Frobenius layer collapses to harmonic sums (the classical formulas, which the construction reproduces); where it cannot, the companion is an Eichler integral of the parametrizing form.

5.3. The census. Run in reverse over a precisely defined universe ($\sim 5 \times 10^4$ hauptmodul/form pairs, fifteen genus-zero levels, stated banks, caps and equivalence relation), the construction manufactures Apéry-like pairs — and finds none new: the pipeline regenerates all ten identified families from their (t, F) data alone (10/10 exact), and every integral three-term hit is a known sporadic or a classical degenerate (Zagier's criterion $a^2 - 4c = 0$ on the nose). *No sixteenth pair in this universe* [VERIFIED: census; [MA]]. Four uncovered directions (generalized-eta/ Γ_1 , Fricke hauptmoduls, order-4, levels > 25) are stated and reusable.

5.4. **The state of the fifteen, in one table.** Statuses per the standing rules ($\dagger$ = identification at discovery grade, coefficientwise to the stated order; theorem-grade rows say so).

family	r	uniformization	companion formula	jet-reach	congruence
A (Franel)	2	level $6^\dagger$	harmonic (known) + Thm 5.3	yes	wt-1 law
B	2	$b(-q)$, lvl $36/9^\dagger$	Thm 5.3, first	no	wt-1 law
C	2	identified †	Thm 5.3, first	—	wt-1 law
D	2	$\Gamma_1(5)$ -type †	harmonic (verified) + Thm	yes	mod-5 rule
E	2	identified † , χ_{-4}	harmonic-twisted (known) + Thm	—	wt-1 law
F	2	identified †	Thm 5.3, first	—	wt-1 law
α (Domb)	3	lvl $12^\dagger$; Sym^2 [PROVED]	harmonic (verified) + Thm	no	$A(p) \equiv b_p$
γ (Apéry)	3	$\Gamma_0(6)$; Sym^2 [PROVED]	harmonic [PROVED] + Thm	—	$A(p) \equiv b_p$
δ	3	lvl 12, mod one hyp.; Sym^2 [PROVED]	Thm 5.3, first	no	$A(p) \equiv b_p$
ε	3	lvl $8^\dagger$; Sym^2 [PROVED]	harmonic (verified) + Thm	no	$A(p) \equiv b_p$
ζ	3	$\Gamma_0(9)$ [PROVED] ; Sym^2 [PROVED]	Thm 5.3, first	no	$A(p) \equiv b_p$; $e=1$
η	3	lvl $20^\dagger$, Sym^2 [PROVED]	Thm 5.3, first	—	$A(p) \equiv b_p$
s_7, s_{10}, s_{18}	2	integral, unidentified	s_{10} known; s_7 known; s_{18} first	—	$A(p) \equiv b_p$ (s 's)

(“jet-reach”: the discriminating-scan verdicts of §4; dashes were not run. Boldface marks the campaign’s additions.)

6. ARITHMETIC SHADOWS

Weight dichotomy. All nine modular-weight-2 families satisfy the classical $A(p) \equiv b_p \pmod{p}$ at every tested prime $p \leq 23$ [VERIFIED: eps51], including the unidentified Cooper three.

A new weight-one law. The six weight-one families fail the classical congruence with an exactly structured defect:

$$A(p) \equiv b_p - b \cdot \chi_N(p) \pmod{p},$$

b the recurrence’s constant parameter and χ_N the quadratic character of the *level* (not the family: Franel has trivial family character and a χ_{-3} level defect); **D** follows its own mod-5 rule. Fitted on $p \leq 23$, the law *passes all sixteen held-out primes* $29 \leq p \leq 97$ for every family [VERIFIED: eps57_heldout] — the referee’s bar for “credible arithmetic evidence,” met. The mechanism (Atkin–Swinnerton-Dyer structure of the identified forms; for ζ the twist exponent equals the nebentypus value at the discriminating primes) is stated as a conjecture with ζ — where the form is a theorem — as the first target ([MA]).

The half-Apéry numbers ([HA]). The symmetric-square theorem gives Apéry’s numbers a square root, $u = \sqrt{y_0}$, and $\tilde{u}_n := 4^n [t^n]u$ is an integer sequence: 1, 10, 534, 40900, 3672550, ... Its integrality, though, is *universal* — $\sqrt{f(4t)} \in \mathbb{Z}[[t]]$ for any integral f (a two-line Catalan-number lemma) — and the programme’s earlier reading of root-integrality as an arithmetic signal (including for the $\zeta(5)$ row) was corrected accordingly, in print. The genuine treasures sit one layer deeper: $v_2(\tilde{u}_n) = s_2(n)$, the binary digit sum [VERIFIED: $n \leq 200$]; and the *split-prime Lucas law*

$$\tilde{u}_{ap+r} \equiv \tilde{u}_a \tilde{u}_r \pmod{p} \iff \left(\frac{-6}{p}\right) = 1 \text{ or } p \in \{2, 3\}$$

— a conjecture verified at fourteen primes with perfect correlation, predicted from four and confirmed at ten — with discriminant -24 exactly the quadratic datum of the weight-one eta identity $u(t(q)) = (\eta_2 \eta_3)^{7/2} / (\eta_1 \eta_6)^{5/2}$. The $\zeta(5)$ root sequences share the 2-adic law and have *no* Lucas law: the split-prime congruence is a fingerprint of the weight-one world, not of root-taking. A small object, found by accident, deflated honestly, and still carrying a class field.

7. THE $\zeta(5)$ FRONTIER

What the campaign established about the Brown–Zudilin family itself, and what remains:

Local signatures [PROVED]: the order-9 generating operator has $t = 0$ exponents $0^{(5)}, \frac{1}{2}$, and an irrational cubic — a five-dimensional unipotent block, the local signature matching weight 5.

Arithmetic signatures [VERIFIED: q^{26}/q^{32}]: the block's mirror map and Yukawa-type datum are integral at scale one ($t(q) = q - 94q^2 - \dots$; $K(q) = 87q + 33895q^2 + \dots$, with $K_1 = 87 = 4P_1$). Root-integrality is *withdrawn* as evidence (universal; [HA]). The q -normal form $\theta_q^5 + K_3\theta_q^3 + \dots$ is displayed in [MA]; $K \not\equiv 0$ is precisely the obstruction to bare inversion at $r = 5$, and K 's gauge-invariance is honestly [OPEN].

Sharp negatives [EXCLUDED]: no order-5 right factor of t -degree ≤ 20 splits the block globally; the naive Dwork congruence fails (three Frobenius slopes; the slope-refined form is the open target); the naive elliptic-style row avatars have growing denominators — calibrated against the same pipeline recovering Beukers' weight-4 Eisenstein combination exactly for Apéry.

The two roads to the bridge. (i) The mirror road: Conjecture ([MA]) — in the block's normal-form coordinates the rows $\hat{P}_n, P_n$ are Eichler-type objects in $F, K, \hat{g}_3, \dots$, and (BRIDGE) is the t -shadow of a q -side identity; the first computation (the *proved* middle row against $\{F, K, \hat{g}_3\}$, with the K -corrected variation-of-parameters inversion) is in progress at the time of writing, as is a systematic denominator harvest from the companion formula (eps58, eps59). (ii) The variational road: the tower of §3, whose remaining one-time cost is the parametric certificate, with Brown's contiguity matrices as template.

Calibration, stated plainly. None of this touches the worthiness exponent. The linear forms decay like $e^{-2.47n}$; the denominators grow like e^{5n} ; Brown–Zudilin's 0.86 is best known and their analysis caps known constructions short of irrationality. The campaign improved the *arithmetic* side — companions, congruences, and the denominator machinery those feed — which is the only known lever on the exponent, and a lever with a hard ceiling. The path to the irrationality of $\zeta(5)$ remains unopened; anyone reporting otherwise is reporting scenery.

8. CONSOLIDATED OPEN PROBLEMS

Deduplicated across all memos, ranked by value over distance; technical homes in brackets.

- P1.** Complete δ 's parametrization proof (one Atkin–Lehner degree-one hypothesis); extend the Ligozat–Sturm route to the remaining eleven identifications. [[MA]; DELTA_MODULAR_THEOREM.md]
- P2.** Test the mirror form of the bridge on the proved middle row: the K -corrected inversion of the five-block; then P_n . [[MA] P4; eps59 in progress]
- P3.** The denominator harvest: derive and prove the bound the Eichler formula implies; compare with the observed sharp denominators of all fifteen companions; state the first denominator theorems for the seven. [eps58 in progress]
- P4.** Prove the weight-one law from the Eisenstein structure of the identified forms; then the ASD mechanism for ζ . [[MA] P2]
- P5.** The variational tower's parametric certificate ($M_{0,8}$ contiguity matrices); or the B_5 null representative with safe jet orders. [[VA]; VARIATIONAL_TOWER.md]
- P6.** Certify the second ζ annihilation law; mine the quotient alphabet. [[MA] P3]
- P7.** Prove the split-prime Lucas law and the $v_2 = s_2$ law for the half-Apéry numbers; identify the inert-prime two-term structure; check the OEIS. [[HA]]
- P8.** Supply $\hat{g}_3, \hat{g}_4$ and K 's gauge-invariance; the slope-refined Dwork test. [[MA] P4]
- P9.** Extend the census in its four uncovered directions; identify Cooper's three. [[MA] P5, P6]
- P10.** The bridge (BRIDGE) itself: the 231-monomial residual by a genuinely global mechanism (uncapped telescoping; the two roads of §7). [BRIDGE_CAMPAIN memo]

APPENDIX A. PROVENANCE: THE COMPLETE MAP

Papers.

tag	path	content
[MA]	modular_anchors	the theory (post-referee): inversion, Sym^2 , all- n companion, identifications, scoped negatives, census, conjectural tier
[VA]	variational_anchor	coupling lemma, curve-blindness, variational anchor and tower
[EA]	expository_apery	pedagogical walkthrough (Apéry), with verification script and Sym^2 lemma
[HA]	half_apery	the half-Apéry numbers

(All under `papers_out/`.)

Memos. `work/`: `BRIDGE_CAMPAIGN_2026-08-03.md` (the sieges); `CURVE_BLINDNESS.md`; `EPS_DISCOVERY.md`; `VARIATIONAL_TOWER.md`; `BROWN26_MINING.md`; `MODULAR_COACTION_PROBE.md` (Parts A–D); `SPORADIC_MODULAR_DIC`; `MODULAR_FACTORY.md`; `DELTA_MODULAR_THEOREM.md`; `Z5_MODULARITY_PROBE.md` (with correction log); `SOL_REVIEW_MODULAR_ANCHORS.md` (the referee report, the point-by-point response, and the standing no-silent-upgrade log); `z5eps/eps45_zeta_kernel.md`.

Scripts. `work/z5eps/`: `eps26–eps33` (sieges, residual, saturation); `eps40–eps42` (realizability; curve scans); `eps43–eps44` (discovery instrument; exhaustion); `eps45` (the proved annihilation law); `eps46–eps47` (variational anchor and tower); `eps48–eps49` (nome instrument; ζ companion; Borwein/ASD); `eps50`, `eps53` (the $\zeta(5)$ mirror probes); `eps51` (dictionary and ASD sweep); `eps52` (Eichler construction, all fifteen); `eps54` (the discriminating experiments); `eps55` (census; δ theorem); `eps56` (half-Apéry); `eps57` (Sym^2 identities, Sturm, held-out primes, saturation); `eps58`, `eps59` (denominator harvest and row hunt; in progress). All exact arithmetic is `Fraction`-based; every modular negative is certified at two unrelated primes with the transfer lemma of [MA]; every positive control is stated alongside the negatives it guards.

External review. The papers [EA] and [MA] each received a hostile referee report from an external assistant (“Sol”); both reports were correct in their central criticisms, both were implemented in full (one with a disagreement count of zero), and both made the mathematics stronger — the symmetric-square theorem exists because a referee refused to accept MUM in its place.

REFERENCES

- [1] *Modular anchors for Apéry-like companions*, `papers_out/modular_anchors`, this programme, 2026.
- [2] *Variational anchors for Apéry-like families*, `papers_out/variational_anchor`, this programme, 2026.
- [3] *The companion by the nome*, `papers_out/expository_apery`, this programme, 2026.
- [4] *The half-Apéry numbers*, `papers_out/half_apery`, this programme, 2026.
- [5] *The fifteen sporadic Apéry-like pairs*, `papers_out/sporadics`, this programme, 2026.
- [6] F. Brown and W. Zudilin, *On cellular rational approximations to $\zeta(5)$* , 2022–2026.
- [7] D. Zagier, *Integral solutions of Apéry-like recurrence equations*, 2009.
- [8] F. Beukers, *Irrationality proofs using modular forms*, 1987.